RAN SOLUTION PLAYBOOK

2025

TABLE OF CONTENT

RAN Solution Playbook

- **Coverage Use Cases - New Infra**
  - New Infra Use Case-1 New Infra High Population with TA < 2K m
  - New Infra Use Case-2 New Infra High Population with Far TA
  - New Infra Use Case-3 New Infra Off Grid (no PLN)
    - 3.1 Regular Infra
    - 3.2 LCI
  - New Infra Use Case-4 New Infra Bad Grid (PLN 12H)
    - 4.1 Regular Infra
    - 4.2 LCI
  - New Infra Use Case-5 New Infra Corporate with ultra high capacity
  - New Infra Use Case-6 Narrow & Long Coverage
- **Coverage Use Cases – Non-New Infra**
  - Non-New Infra Use Case-1 New Coverage with low Population
    - 1.1 Low Population with Fronthaul
    - 1.2 Low Population with IPMW/OFDM
    - 1.3 Low Population with no transport available
  - Non-New Infra Use Case-2 Indoor Coverage (Inbuilding)
  - Non-New Infra Use Case-3 Indoor Coverage (High Rise Building)
  - Non-New Infra Use Case-4 Micro Coverage (HetNet)
  - Non-New Infra Use Case-5 Toll Road
  - Non-New Infra Use Case-6 Sea Coverage
  - Non-New Infra Use Case-7 High Population Wide Coverage
  - Non-New Infra Use Case-8 High Population Wide Coverage UL Guarantee

TABLE OF CONTENT

→**Capacity Opportunistic**

→Opportunistic Use Case-1 Power Limitation

→Opportunistic Use Case-2 Infra Limitation

→Opportunistic Use Case-3 Full Config Solution (FMM/Triple Beam)

→Opportunistic Use Case-4 Over Quota Solution

→**B2B Corporate Use Cases**

→B2B Corporate Use Case-1 Private Network Enterprise

→**Support Event**

→Support Event Use Case-1 Small Event

→Support Event Use Case-2 Mid Event

→Support Event Use Case-3 Large Event

→Support Event Use Case-4 Indoor Event

RAN Product Catalogue

→Existing Product till 2024 .....

→New Products in 2025 .....

COVERAGE USE CASE:

Telkomsel

NEW INFRA HIGH POPULATION WITH TA < 2KM

Background

Many new Infra single band existing become high util in less than three month. Second band took long time deployment more than three month due to PO process and hardware readiness.

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Low OPEX with simplified hardware

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

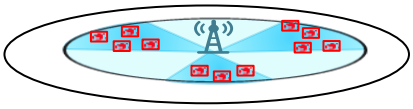
Solution

- Simplified New Infra with Dual Band Hardware Ready
- Low power consumption and tower space requirement

Scenario

New Infra L1800+L2100

Site Radius < 2km



B2B Corporate

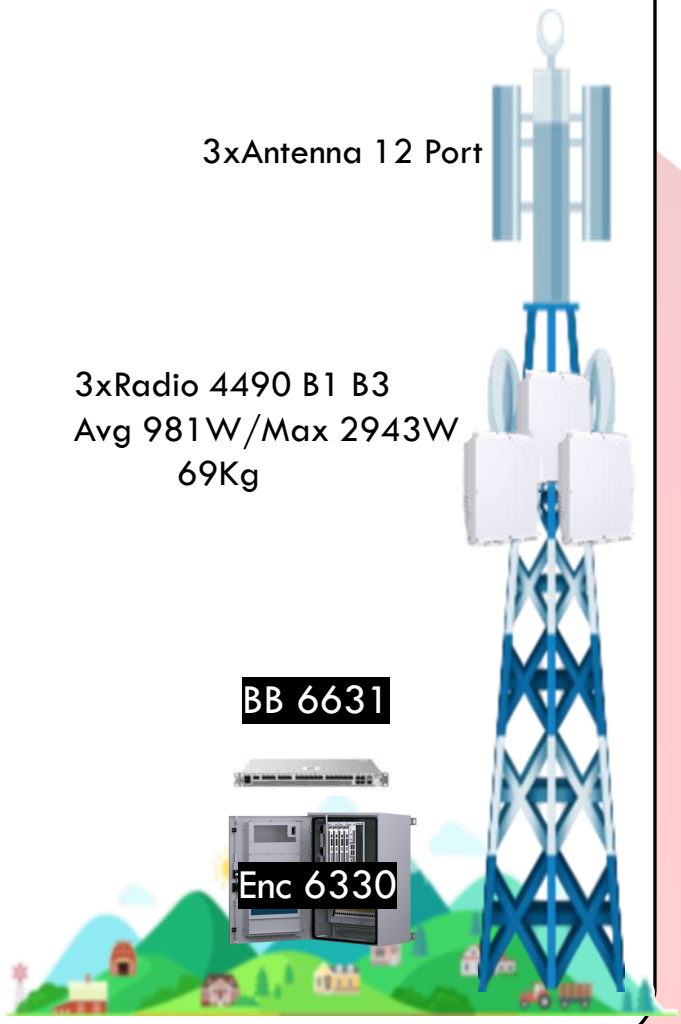
Sub Urban

- High Population Area
- Estimated user population >3000

LTE 1800	**3 x20MHz**
LTE 2100	3 x20MHz (HW)
DCS 1800	3x3 Trx

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC

Site Configuration



COVERAGE USE CASE:

Telkomsel

NEW INFRA HIGH POPULATION WITH TA < 2KM

New : Radio 6626 B1 B3

Background

Many new Infra single band existing become high util in less than three month. Second band took long time deployment more than three month due to PO process and hardware readiness.

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Low OPEX with simplified hardware

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

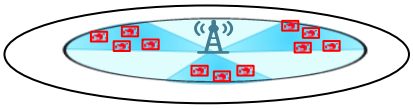
Solution

- Simplified New Infra with Dual Band Hardware Ready
- Low power consumption and tower space requirement

Scenario

New Infra L1800+L2100

Site Radius < 2km

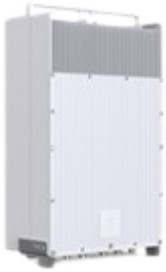


B2B Corporate

Sub Urban

- High Population Area
- Estimated user population >3000

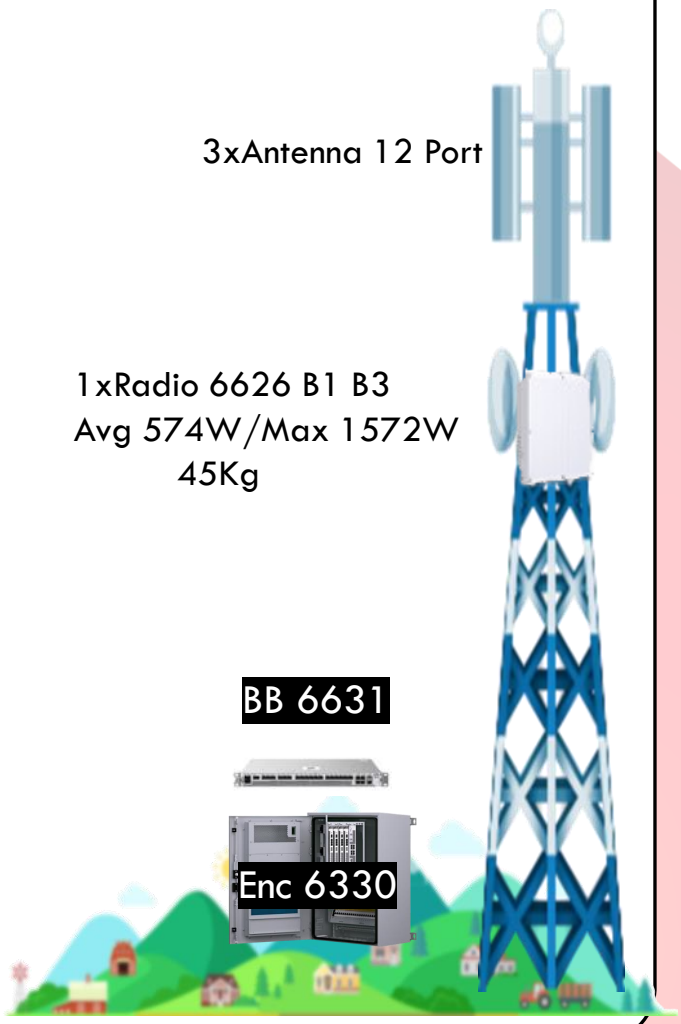
LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)
DCS 1800	3x3 Trx



Radio 6626 B1 B3

- 1800+2100 MHz
- 2G/4G/5G/Nb-IoT
- 6x120 Watt
- 3 x 2T2R
- 45kg | -48 VDC

Site Configuration



COVERAGE USE CASE:

Telkomsel

NEW INFRA HIGH POPULATION WITH FAR TA

Background

New Infra single band become high util in less than 3-month, need New NE for additional capacity.

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

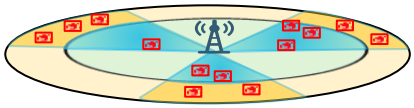
Solution

- New Infra with Dual Band Hardware Ready for Low Band
- Low power consumption and tower space requirement

Scenario

New Infra L900+L700

Site Radius 1km ~ 3km

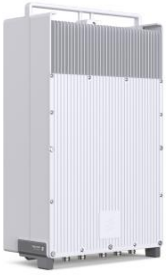


Rural

B2B Corporate

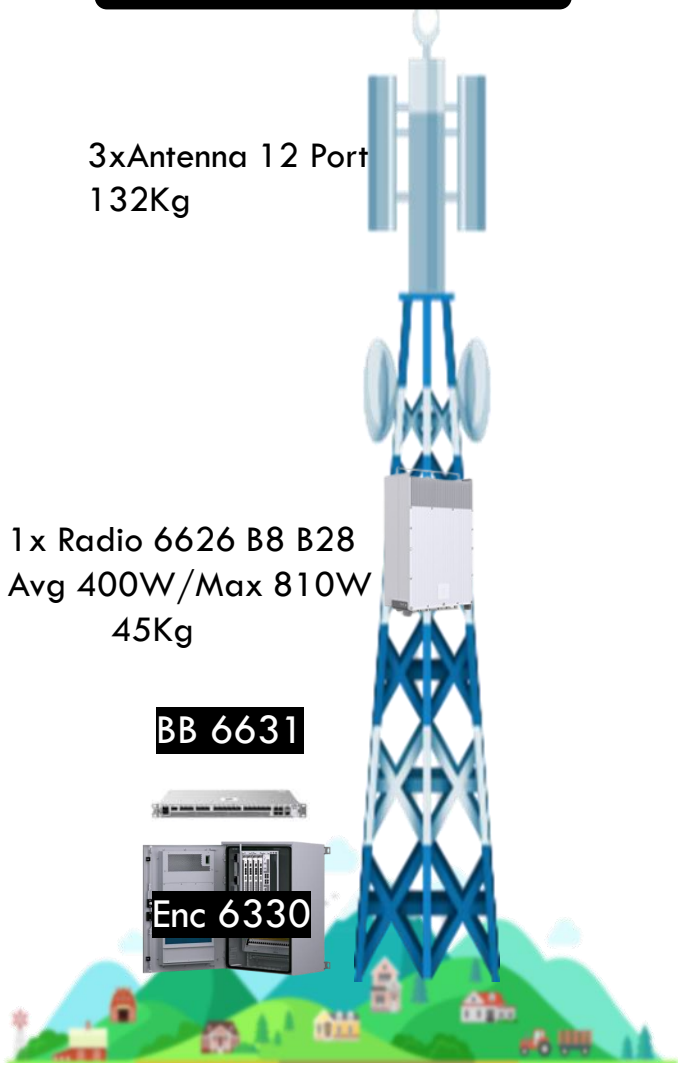
- Low Population Area
- Estimated user population 1000-3000

LTE 900	**3 x10MHz**
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
NEW INFRA OFF GRID (NO PLN) - REGULAR INFRA

Telkomsel

Background

New Infra Without PLN with regular infra, need 2nd NE for additional Capacity

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Optimize Solar Panel System

Transport & Power & Infra

- Fiber Optic/Radio IP available
- No PLN
- Tower height 30 meter or more

Solution

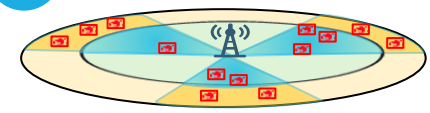
- Simplified New Infra with Dual Band Hardware Ready with Low Power Consumption

Scenario

New Infra L900+L700

1

Site Radius 1km ~ 3km

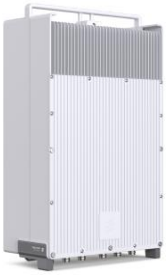


Rural

B2B Corporate

- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

New Infra L1800+L2100

2

Site Radius < 2km

B2B Corporate

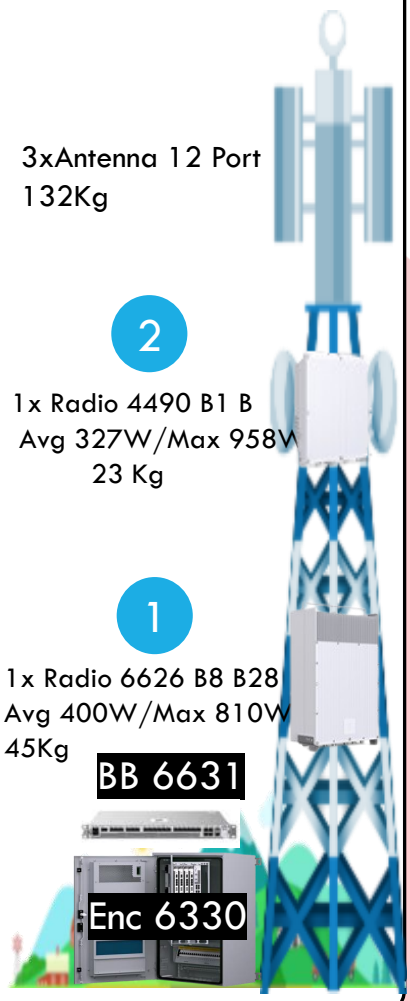
Sub Urban

- High Population Area
- Estimated user population >3000

LTE 1800	2 x20MHz
LTE 2100	2 x20MHz (HW)
DCS 1800	2x3 Trx

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
NEW INFRA OFF GRID (NO PLN) - LCI

Telkomsel

Background

New Low Cost Infra (LCI) without PLN, need 2nd NE for additional Capacity, Coverage is Near to Medium.

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Optimize Solar Panel System

Transport & Power & Infra

- Fiber Optic/Radio IP available
- No PLN
- LCI

Solution

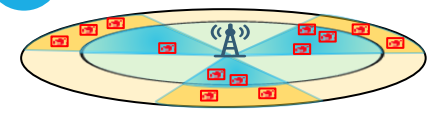
- Simplified New Infra with Dual Band Hardware Ready with Low Power Consumption

Scenario

New Infra L900+L700

1

Site Radius 1km ~ 3km

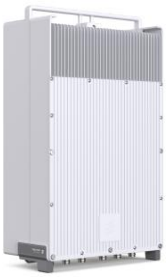


Rural

B2B Corporate

- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

New Infra L1800+L2100

2

Site Radius < 2km

B2B Corporate

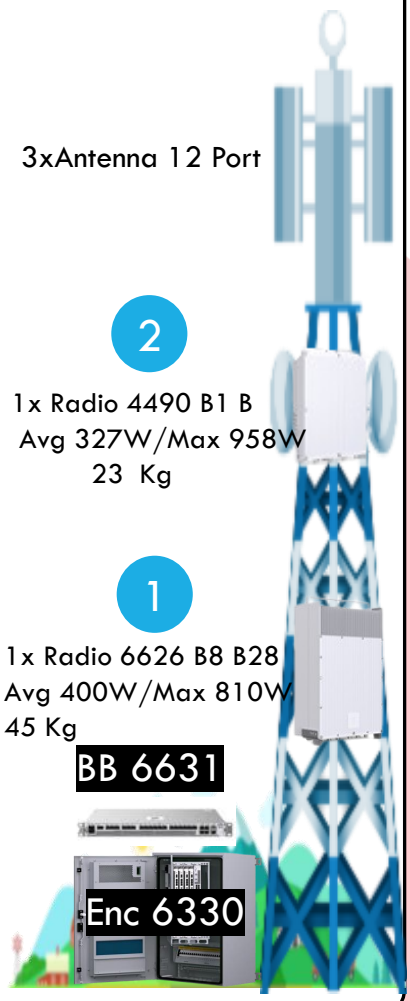
Sub Urban

- High Population Area
- Estimated user population >3000

LTE 1800	2 x20MHz
LTE 2100	2 x20MHz (HW)
DCS 1800	2x3 Trx

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
NEW INFRA BAD GRID (PLN 12H) - REGULAR INFRA

Telkomsel

Background

New Infra with limited PLN availability (12H) and regular infra, need 2nd NE for additional Capacity.

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Optimize existing Power Supply

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN 12H
- Tower height 30 meter or more

Solution

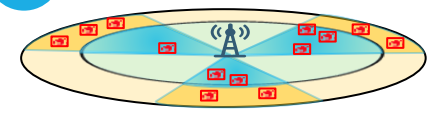
- Simplified New Infra with Dual Band Hardware Ready with Low Power Consumption

Scenario

New Infra L900+L700

1

Site Radius 1km ~ 3km

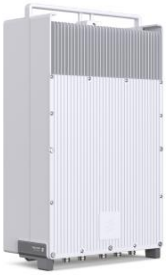


Rural

B2B Corporate

- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

New Infra L1800+L2100

2

Site Radius < 2km

B2B Corporate

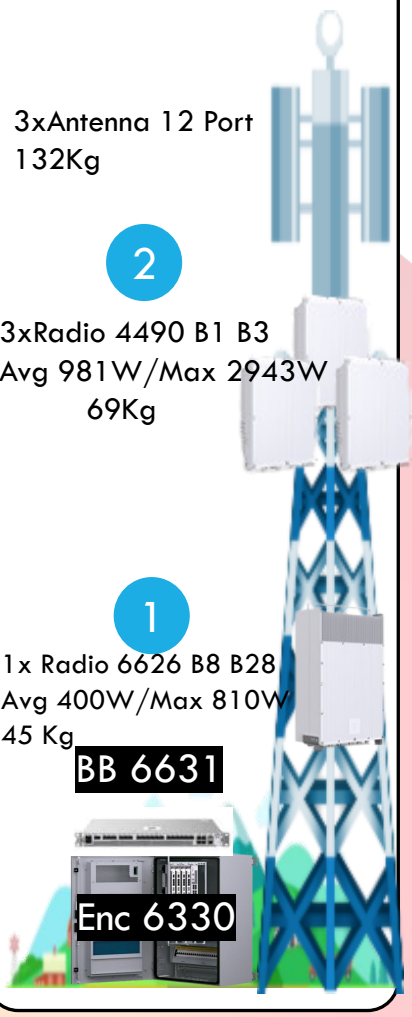
Sub Urban

- High Population Area
- Estimated user population >3000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)
DCS 1800	3x3 Trx

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
NEW INFRA BAD GRID (PLN 12H) - LCI

Telkomsel

Background

New Low Cost Infra (LCI) with limited PLN availability (12H) and regular infra, need 2nd NE for additional Capacity, Coverage is Near to Medium.

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Optimize existing Power Supply

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN 12H
- LCI

Solution

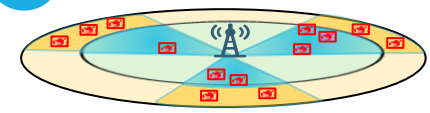
- Simplified New Infra with Dual Band Hardware Ready with Low Power Consumption

Scenario

New Infra L900+L700

1

Site Radius 1km ~ 3km

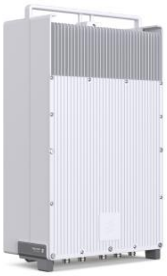


Rural

B2B Corporate

- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

New Infra L1800+L2100

2

Site Radius < 2km

B2B Corporate

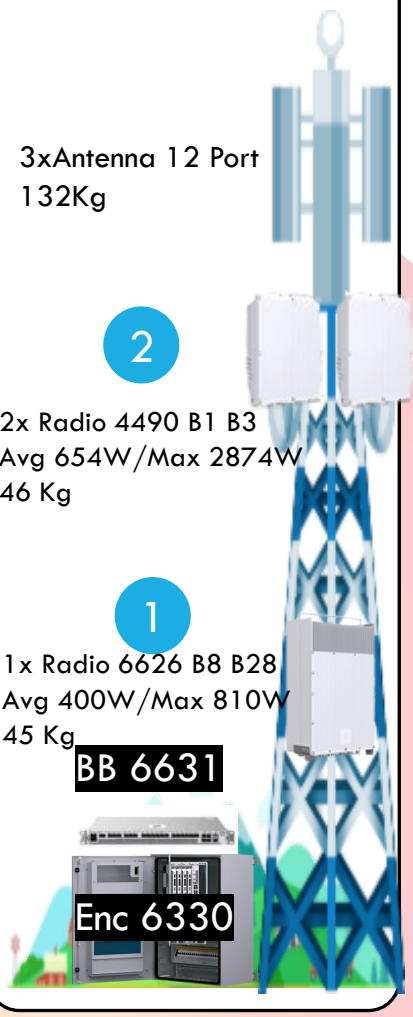
Sub Urban

- High Population Area
- Estimated user population >3000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)
DCS 1800	3x3 Trx

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
32kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
NEW INFRA CORPORATE WITH ULTRA HIGH CAPACITY

Telkomsel

Background

New Infra with high dense population in corporate area need ultra high capacity

Objective

- Hardware Readiness in first deployment
- Fast second band deployment
- Boost productivity & revenue
- Maximize payload & revenue, avoid loss due to long deployment time

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

Solution

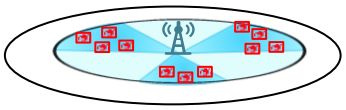
- Ultra-High Capacity New Infra

Scenario

New Infra L1800+L2100+L2300

1

Site Radius < 2km

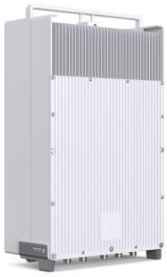


B2B Corporate

Sub Urban

- High Population Area
- Estimated user population >3000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)
LTE 2300	3x50MHz (HW)
DCS 1800	3x3 Trx

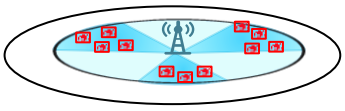


Radio 4466
1800+2100+2300 MHz
2G/4G/5G/Nb-IoT
4x180 Watt
4T4R
45kg | -48 VDC

New Infra L1800+L2100

2

Site Radius < 2km



B2B Corporate

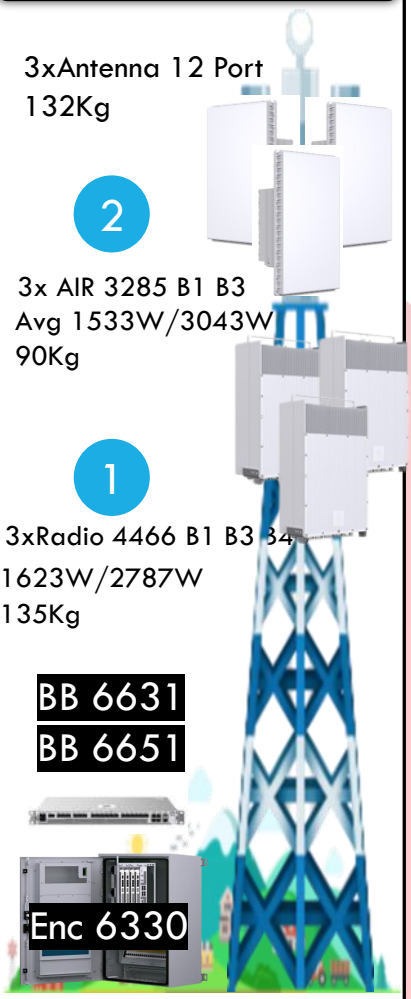
Sub Urban

- High Population Area
- Estimated user population > 3000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)

FDD MM
1800+2100 MHz
4G/5G
320 Watt
32T32R
30kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
NARROW & LONG COVERAGE

Telkomsel

Background

New Infra for narrow and long coverage suitable for toll routes and long distance coverage

Objective

- Connecting Road for contiguous coverage
- Cost-effective solution for low population

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

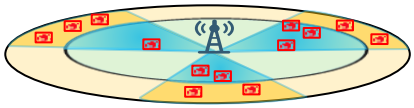
Solution

- New infra low band with high gain antenna

Scenario

New Infra L900+L700

Site Radius 1km ~ 3km

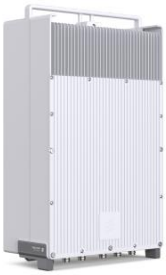


Rural

B2B Corporate

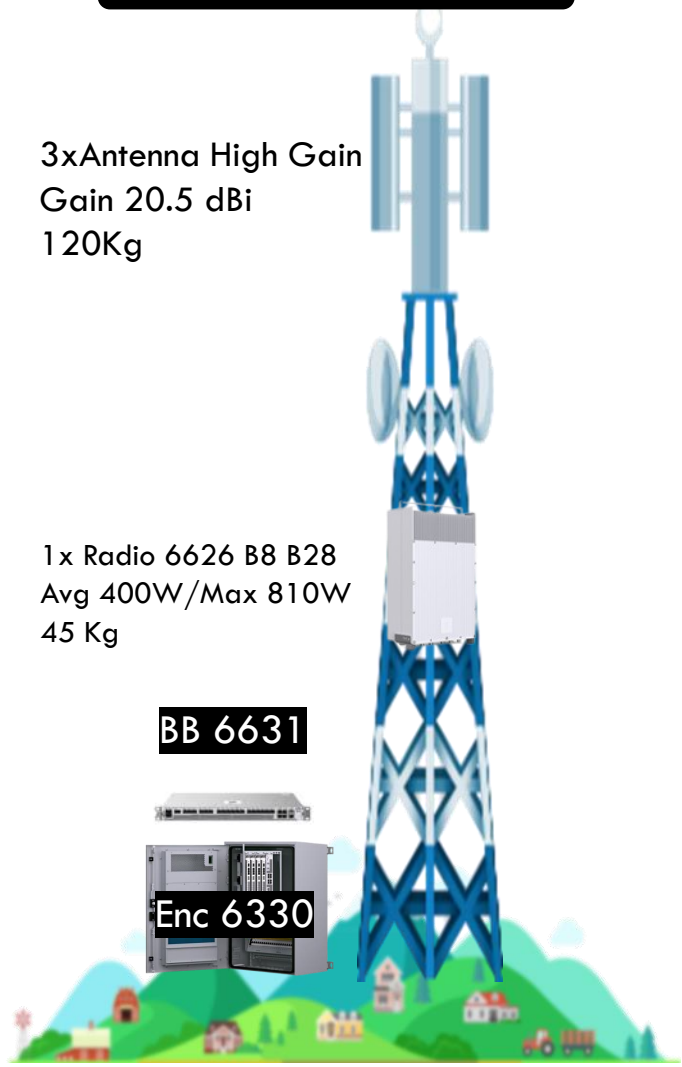
- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
COVERAGE EXTENSION WITH FRONTHAUL

Telkomsel

Background

Coverage extension with optical fronthaul available.

Objective

- Cost effective solution for low population
- Coverage extension to increase productivity & revenue

Transport & Power & Infra

- Fiber Optic available
- PLN available
- Tower height lower than 30 meter/Pole

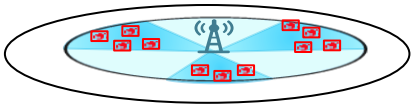
Solution

- One leg/Easy Pole solution

Scenario

Street Macro L1800+L2100

1 Site Radius < 1km



B2B Corporate

Sub Urban

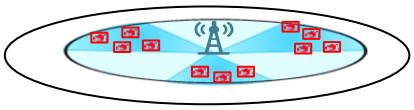
- Mid Population Area
- Estimated user population <1000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)
DCS 1800	3x3 Trx

AIR 4455
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x40 Watt
4T4R
27kg | -48 VDC

RMA L2300

2 Site Radius < 1km



B2B Corporate

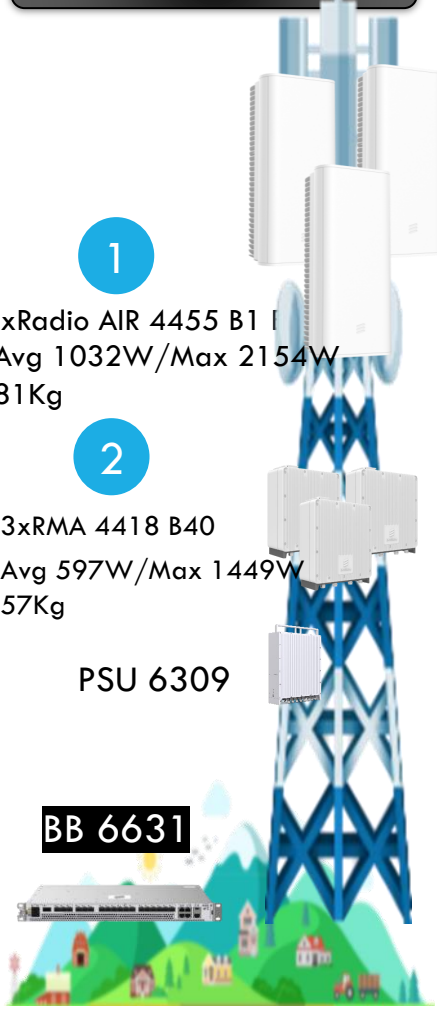
Sub Urban

- Mid Population Area
- Estimated user population <1000

LTE 2300	3 x50MHz

Radio 4418+RMA
2300 MHz
4G/5G/Nb-IoT
4x40 Watt
4T4R
19kg | AC/-48 VDC

Site Configuration



COVERAGE USE CASE:
COVERAGE EXTENSION WITH IPMW/OFDM

Telkomsel

Background

Coverage extension with IPMW/OFDM available.

Objective

- Cost effective solution for low population
- Coverage extension to increase productivity & revenue

Transport & Power & Infra

- Radio IP available
- PLN available
- Tower height 30 meter or more

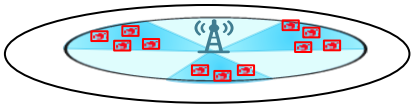
Solution

- Remote BBU solution

Scenario

Street Macro L1800+L2100

1 Site Radius < 1km



B2B Corporate

Sub Urban

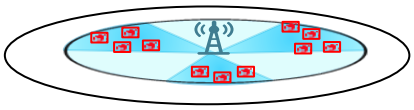
- Mid Population Area
- Estimated user population <1000

LTE 1800	**3 x20MHz**
LTE 2100	3 x20MHz (HW)

AIR 4455
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x40 Watt
4T4R
27kg | -48 VDC

RMA L2300

2 Site Radius < 1km



B2B Corporate

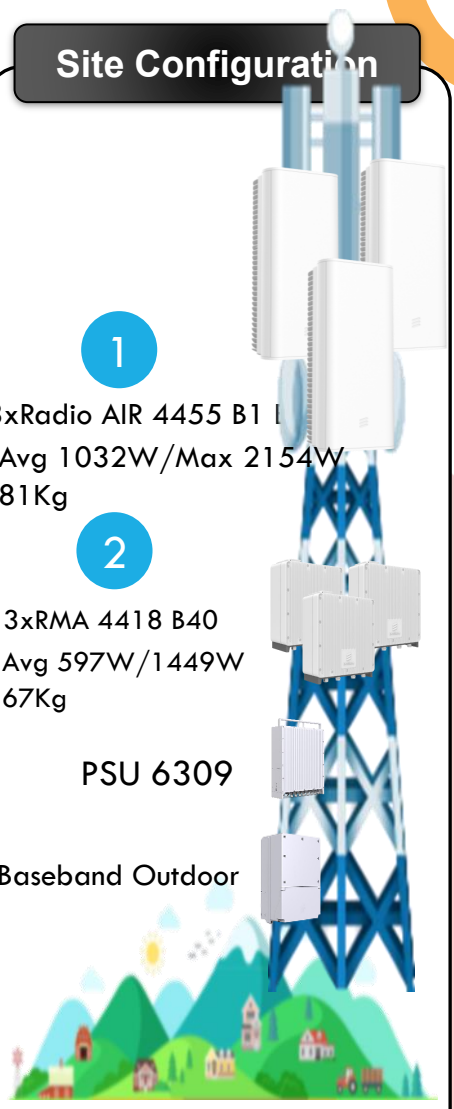
Sub Urban

- Mid Population Area
- Estimated user population <1000

LTE 2300	**3 x50MHz**

Radio 4418+RMA
2300 MHz
4G/5G/Nb-IoT
4x40 Watt
4T4R
19kg | AC/-48 VDC

Site Configuration



1
3xRadio AIR 4455 B1
Avg 1032W/Max 2154W
81Kg

2
3xRMA 4418 B40
Avg 597W/1449W
67Kg

PSU 6309

Baseband Outdoor

COVERAGE USE CASE:
COVERAGE EXTENSION WITH NO TRANSPORT AVAILABLE

Background

Coverage extension with no transport available.

Objective

- Cost effective solution for low population
- Coverage extension to increase productivity & revenue

Transport & Power & Infra

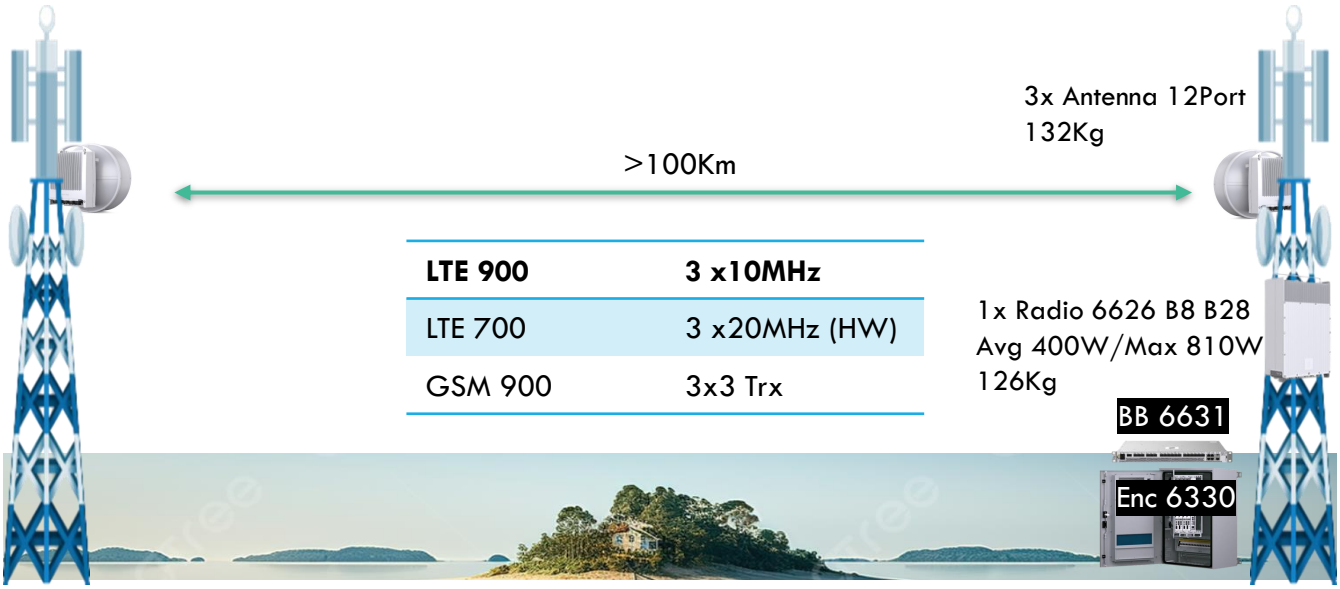
- Fiber Optic/Radio IP not available
- PLN available
- Tower height 30 meter or more

Solution

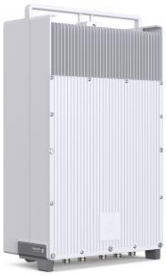
- Alternative transport solution/Relay Solution/Controlled repeater

Scenario

Minilink Long Haul > 100 Km with Radio Low Band Dual band



LTE 900	**3 x10MHz**
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Minilink 6000
6GHz 4x (1+1SD)
3m DP

COVERAGE USE CASE:
COVERAGE EXTENSION WITH NO TRANSPORT AVAILABLE

Background

Coverage extension with no transport available.

Objective

- Cost effective solution for low population
- Coverage extension to increase productivity & revenue

Transport & Power & Infra

- Fiber Optic/Radio IP not available
- PLN available
- Tower height 30 meter or more

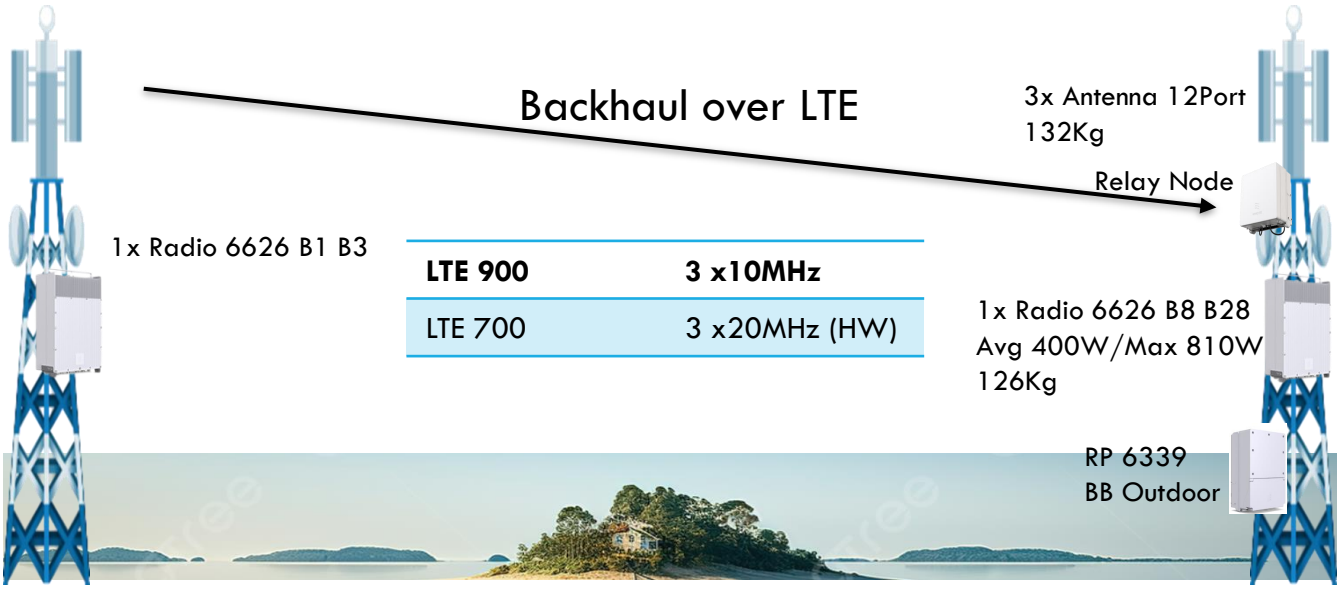
Solution

- Alternative transport solution/Relay Solution/Controlled repeater

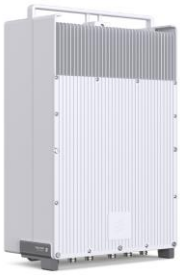
Status : LTE Relay Not Yet FNI

Scenario

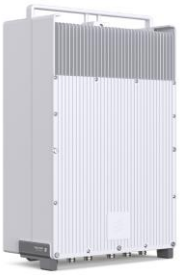
LTE Relay



LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC



Radio 6626
1800+2100 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

RP 6339
- SDR, LTE+NR
- Outdoor,10.5 Kg
- Power AC or DC

COVERAGE USE CASE:
INDOOR COVERAGE (INBUILDING)

Telkomsel

Background

Large building area need new coverage or improve existing coverage, not possible to cover from Macro or existing DAS.

Objective

- Improve Indoor Coverage
- 5G Ready for Indoor
- Boost Productivity & revenue
- Improve indoor user experience

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Indoor

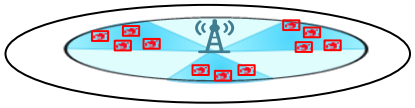
Solution

- Indoor Coverage Solution

Scenario

Indoor L1800+L2100

Site Radius < 300m



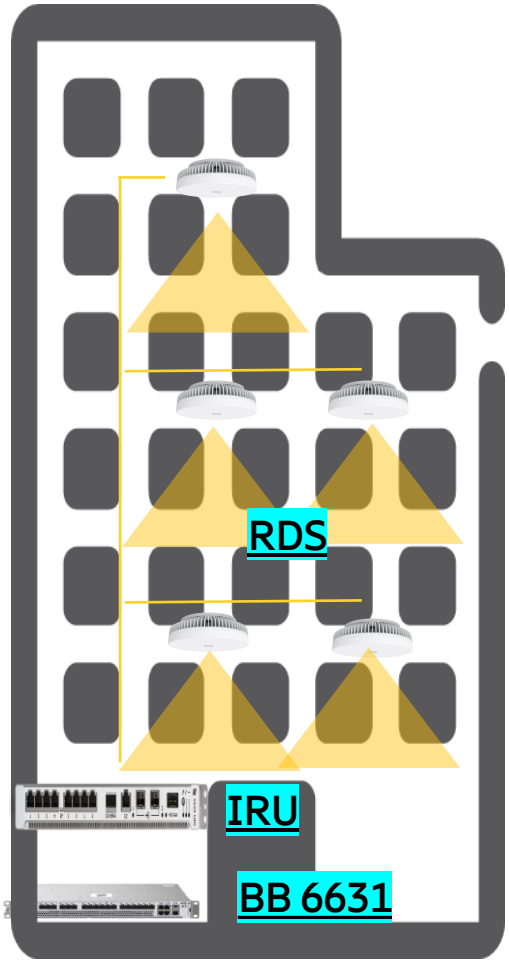
Building **Dense Urban**

- High Population Area
- Estimated user population <1000

LTE 1800	20MHz
LTE 2100	20MHz (HW)
DCS 1800	1 Trx

Radio Dot
1800+2100 MHz
2G/4G/5G
2x125 mWatt
2T2R
0.5kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
INDOOR COVERAGE (HIGH RISE BUILDING)

Telkomsel

Background

High Rise building area need new coverage or improve existing coverage with no possibility to build or modify indoor antenna system.

Objective

- Improve High Rise Building Coverage
- 5G Ready for Indoor
- Boost Productivity & revenue
- Improve indoor user experience

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Roof top/Pole

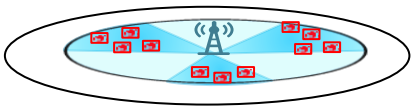
Solution

- Antenna Shooter/TDD Massive MIMO 64TR

Scenario

Massive MIMO L2300

Site Radius < 1km



B2B Corporate

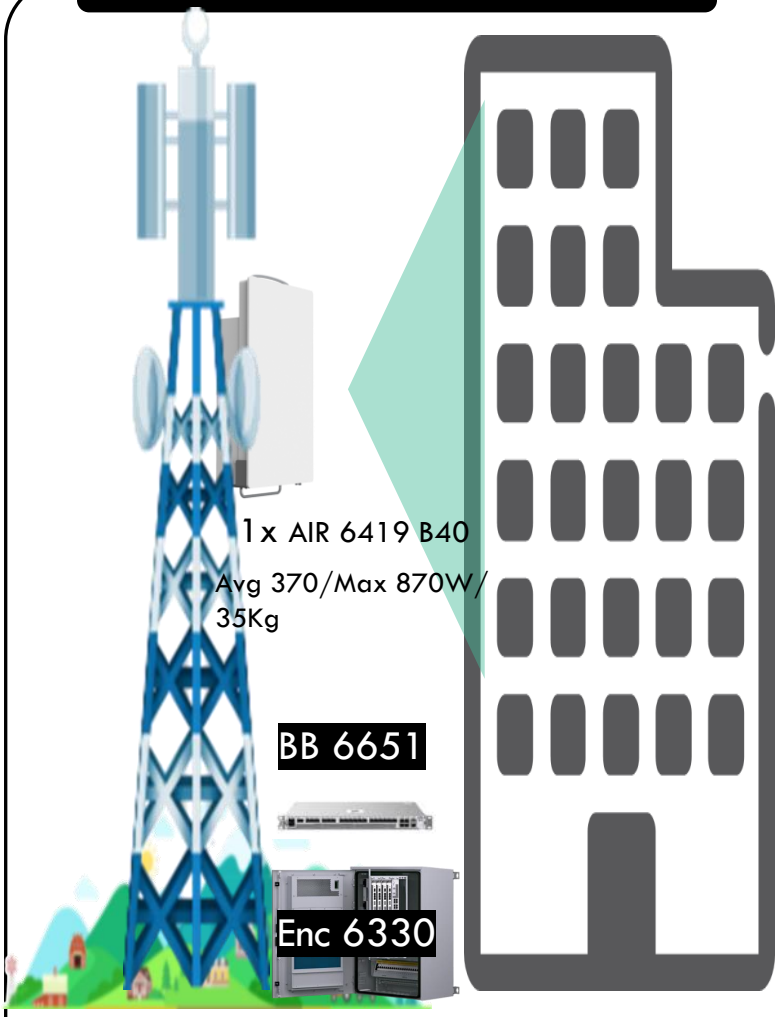
Dense Urban

- High Population Area
- Estimated user population > 3000

LTE 2300	**1 x50MHz**

AIR 6419
2300 MHz
4G/5G/Nb-IoT
320 Watt
64T64R
27kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
MICRO COVERAGE (HETNET)

Telkomsel

Background

Micro cell coverage to fill in blind spot coverage or capacity infill.

Objective

- Reduce coverage blind spot
- Improve User experience
- Increase Productivity & Revenue

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Pole/Wall

Solution

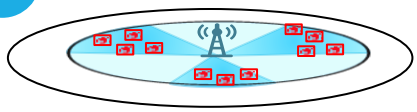
- Micro Cell Coverage/HetNet Solution

Scenario

Micro L1800+L2100

Site Radius < 500m

1



B2B Corporate

Dense Urban

- Mid Population Area
- Estimated user population <1000

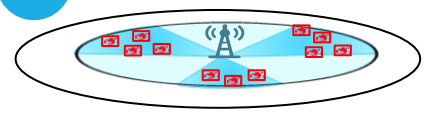
LTE 1800	**1 x20MHz**
LTE 2100	1 x20MHz

Radio 4402 B1
Radio 4402 B3
1800 MHz
4G/5G/Nb-IoT
4x5 Watt
4T4R
10kg | -48 VDC

Micro L2300

Site Radius < 500m

2



B2B Corporate

Dense Urban

- High Population Area
- Estimated user population <1000

LTE 2300	**1 x50MHz**

Radio 4408
2300 MHz
4G/5G/Nb-IoT
4x5 Watt
4T4R
10kg | -48 VDC

Site Configuration

1

Radio 4402 B1

Radio 4402 B3

2

Radio 4408 B40

Avg 43W/Max 93W

10Kg

BB 6631

Enc 6330

WALL

COVERAGE USE CASE:
TOLL ROAD

Telkomsel

Background

Additional sector or coverage extension to cover toll road.

Objective

- Connecting Road for contiguous coverage
- Cost-effective solution for low population

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

Solution

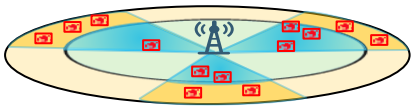
- High Power Low Band RRU + High Gain Antenna

Status : FNI Antenna Not Yet

Scenario

New NE L900+L700

Site Radius 2km ~ 4km

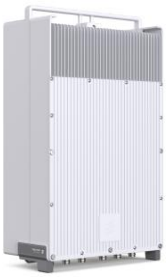


Rural

B2B Corporate

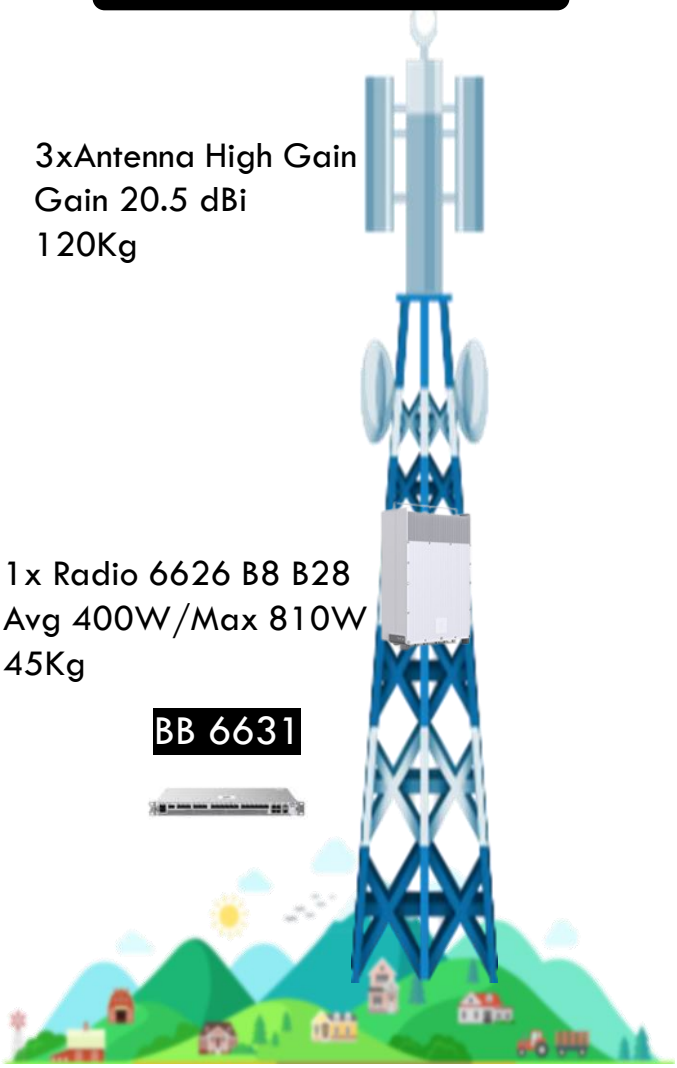
- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
SEA COVERAGE

Telkomsel

Background

There are potential use cases in sea that has no coverage.

Objective

- Improve coverage at sea
- Support connectivity for inter-island, shipping lane, tourism, Offshore RIG
- Cost-effective solution for low population

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 70 meter or more

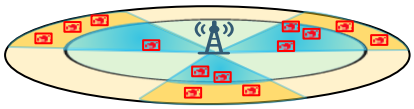
Solution

- High Power Low Band RRU + 4-port Low Band Antenna
- Features Maximum Cell Range Activation

Scenario

New NEL900+L700

Site Radius Up to 100Km

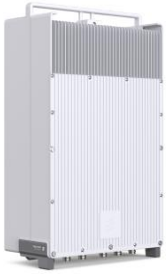


Rural

B2B Corporate

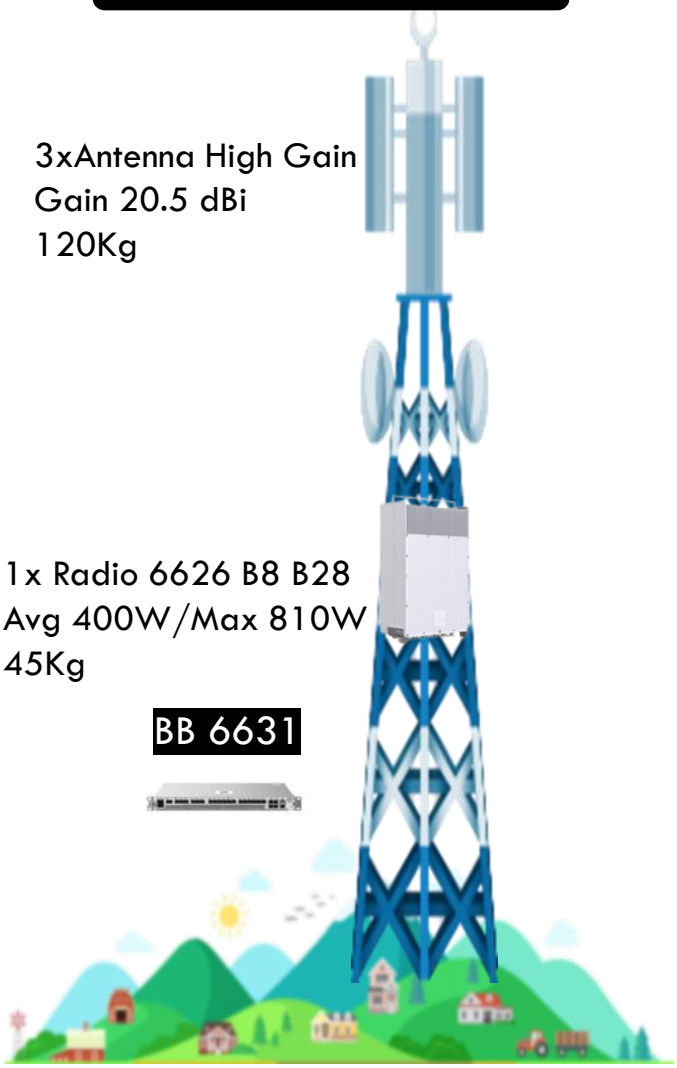
- Low Population Area
- Estimated user population 1000-3000

LTE 900	3 x10MHz
LTE 700	3 x20MHz (HW)
GSM 900	3x3 Trx



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration



COVERAGE USE CASE:
HIGH POPULATION WIDE COVERAGE

Telkomsel

Background

Coverage extension with simplification for wide coverage area without UL guaranteed.

Objective

- Simplify hardware configuration
- Solve wide coverage issue
- Boost Productivity & Revenue
- OPEX saving by reduce equipment at tower

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

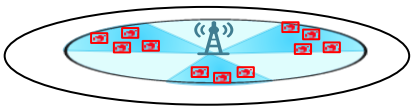
Solution

- TDD Massive MIMO with wide coverage setting

Scenario

New NE L2300

Site Radius < 2km



B2B Corporate

Sub Urban

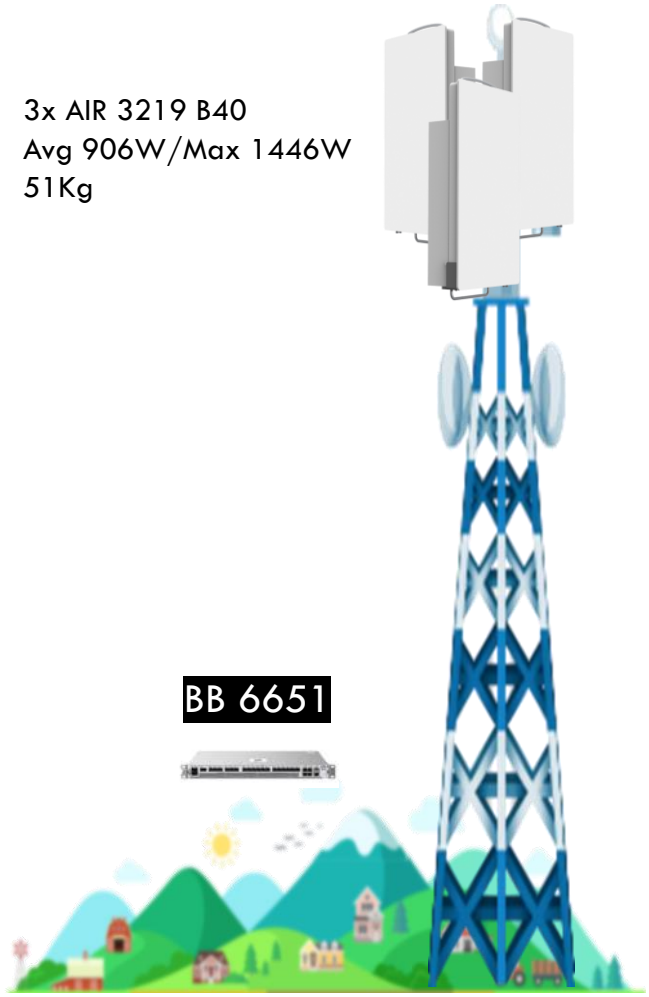
- High Population Area
- Estimated user population > 3000

LTE 2300	**3 x50MHz**

AIR 3219
2300 MHz
4G/5G/Nb-IoT
240 Watt
32T32R
27kg | -48 VDC

Site Configuration

3x AIR 3219 B40
Avg 906W/Max 1446W
51Kg



COVERAGE USE CASE:
HIGH POPULATION WIDE COVERAGE UL GUARANTEE

Telkomsel

Background

Coverage extension with simplification for wide coverage area with UL guaranteed.

Objective

- Simplify hardware configuration
- Solve wide coverage issue
- Boost Productivity & Revenue
- OPEX saving by reduce equipment at tower

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

Solution

- FDD Massive MIMO with wide coverage setting

Status : Not Yet FNI

Scenario

New NE L1800+L2100

Site Radius < 2km

B2B Corporate

Sub Urban

- High Population Area
- Estimated user population > 3000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)

AIR 3285 FDD Massive MIMO 1800+2100 MHz 4G/5G
320 Watt
32T32R
30kg | -48 VDC

Site Configuration

3x AIR 3285 B1 B3
Avg 1533W/3042W
90Kg

CAPACITY OPPORTUNISTIC:
POWER LIMITATION

Telkomsel

Background

Long aging RCI with power limitation due to SPS, PLN availability, need add NE to ease capacity congestion.

Objective

- Add New NE to solve RCI & ease congestion
- Boost Productivity & Revenue
- Optimize existing Power Supply

Transport & Power & Infra

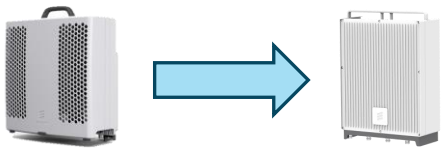
- Fiber Optic/Radio IP available
- PLN limited
- Tower height 30 meter or more

Solution

- Low Power consumption RRU + Baseband

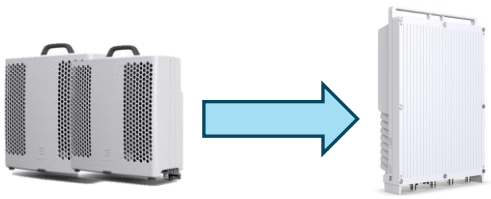
Scenario

Modernization Existing Module



3x RRUS12 B8
1893 W

2x Radio 4499 B8 B28
902 W



3x RRUS12 B1+B3
3747 W

3x Radio 4490 B1 B3
2874 W

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC

Radio 4499
700+900 MHz
2G/4G/5G/Nb-IoT
4x80 Watt
4T4R
32kg | -48 VDC

Site Configuration

3xAntenna 12 Port
132Kg

3x AIR 3219 B40
Avg 906W/Max 1446W
51Kg

3xRadio 4490 B1 B3
Avg 981W/Max 2943W
69Kg

2xRadio 4499 B8 B28
Avg 440W/Max 902W
64 Kg

BB 6631

BB 6651

Enc 6330

CAPACITY OPPORTUNISTIC:
POWER LIMITATION

Telkomsel

Background

Long aging RCI with power limitation due to SPS, PLN availability, need add NE to ease capacity congestion.

Objective

- Add New NE to solve RCI & ease congestion
- Boost Productivity & Revenue
- Optimize existing Power Supply

Transport & Power & Infra

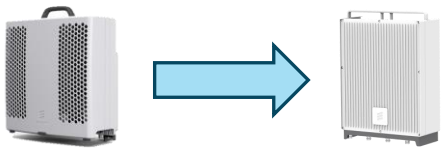
- Fiber Optic/Radio IP available
- PLN limited
- Tower height 30 meter or more

Solution

- Low Power consumption RRU + Baseband 3x2T2R Radio

Scenario

Modernization Existing Module

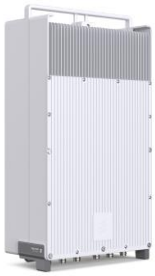


3x RRUS12 B8
1893 W

2x Radio 4499 B8 B28
902 W

3x RRUS12 B1+B3
3747 W

1x Radio 6626 B1 B3
2874 W



Radio 6626 B1 B3
- 1800+2100 MHz
- 2G/4G/5G/Nb-IoT
- 6x120 Watt
- 3 x 2T2R
- 45kg | -48 VDC

Radio 4499
700+900 MHz
2G/4G/5G/Nb-IoT
4x80 Watt
4T4R
32kg | -48 VDC

Site Configuration

3xAntenna 12 Port
132Kg

3x AIR 3219 B40
Avg 906W/Max 1446W
51Kg

1xRadio 6626 B1 B3
Avg 574W/Max 1572W
45Kg

2xRadio 4499 B8 B28
Avg 440W/Max 902W
64 Kg

BB 6631

BB 6651

Enc 6330

CAPACITY OPPORTUNISTIC:
INFRA LIMITATION

Telkomsel

Background

Long aging RCI with infra limitation due to guyed mast, tower space not available, MCR, need add NE to ease capacity congestion.

Objective

- Add New NE to solve RCI & ease congestion
- Boost Productivity & Revenue
- Optimize existing Infra

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Guyed Mast/Combat/Pole

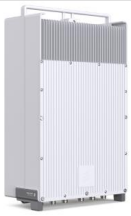
Solution

- Simplified Radio Solution

Scenario

Modernization Existing Module

3x RRUS12 B8
1893 W

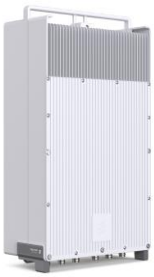


1x Radio 6626 B8 B28
810 W

3x RRUS12 B1+B3
3747 W

3x Radio 4490 B1 B3
2874 W

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration

3xAntenna 12 Port
132Kg

3x AIR 3219 B40
Avg 906W/Max 1446W
51Kg

3xRadio 4490 B1 B3
Avg 981W/Max 2943W
69Kg

1x Radio 6626 B8 B28
Avg 400W/Max 810W
45Kg

BB 6631

BB 6651

Enc 6330

CAPACITY OPPORTUNISTIC:
INFRA LIMITATION

Telkomsel

Background

Long aging RCI with infra limitation due to guyed mast, tower space not available, MCR, need add NE to ease capacity congestion.

Objective

- Add New NE to solve RCI & ease congestion
- Boost Productivity & Revenue
- Optimize existing Infra

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Guyed Mast/Combat/Pole

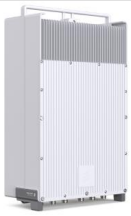
Solution

- Simplified Radio Solution with 3x2T2R Radio

Scenario

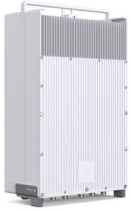
Modernization Existing Module

3x RRUS12 B8
1893 W

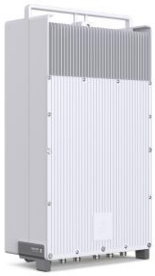


1x Radio 6626 B8 B28
810 W

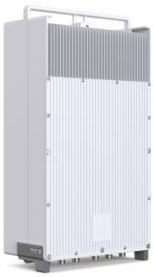
3x RRUS12 B1+B3
3747 W



1x Radio 6626 B1 B3
2874 W



Radio 6626 B1 B3
- 1800+2100 MHz
- 2G/4G/5G/Nb-IoT
- 6x120 Watt
- 3 x 2T2R
- 45kg | -48 VDC



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration

3xAntenna 12 Port
132Kg

3x AIR 3219 B40
Avg 906W/Max 1446W
51Kg

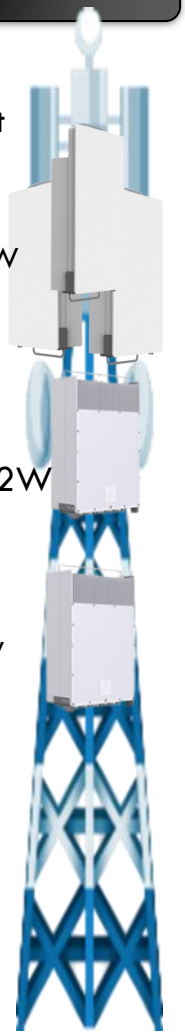
1xRadio 6626 B1 B3
Avg 574W/Max 1572W
45Kg

1x Radio 6626 B8 B28
Avg 400W/Max 810W
45Kg

BB 6631

BB 6651

Enc 6330



CAPACITY OPPORTUNISTIC:
SITE SIMPLIFICATION

Telkomsel

Background

Long aging RCI with infra limitation due to guyed mast, tower space not available, MCR, need add NE to ease capacity congestion.

Objective

- Add New NE to solve RCI & ease congestion
- Boost Productivity & Revenue
- Optimize existing Infra

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Guyed Mast/Combat/Pole

Solution

- Simplified Radio Solution with Interleaved Antenna

Status : Not Yet FNI

Scenario

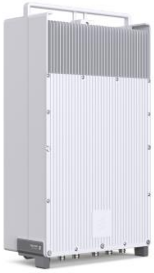
Site Configuration

Interleaved AIR
700+900
1800+2100 (MM)
2300+2600 (MM)
2G/4G/5G
32T32R
120kg | -48 VDC

AIR 3229 TDD
Massive MIMO
2300+2600 MHz
4G/5G
460 Watt
32T32R
30kg | -48 VDC

1x Radio 6626 B8 B28
Avg 400W/Max 810W
45Kg

AIR 3285 FDD
Massive MIMO
1800+2100 MHz
4G/5G
320 Watt
32T32R
30kg | -48 VDC



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

BB 6631

BB 6651

CAPACITY OPPORTUNISTIC:
FULL CONFIG SOLUTION (FMM/TRIPLE BEAM)

Telkomsel

Background

RCI or High Utilization with existing already full band, need capacity upgrade with higher multi-antenna solution.

Objective

- Effective solution for full band configuration
- Boost Productivity & Revenue
- Ease capacity congestion & improve user experience

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

Solution

- FDD Massive MIMO/Triple beam solution

Scenario

Multisector to Triple Beam

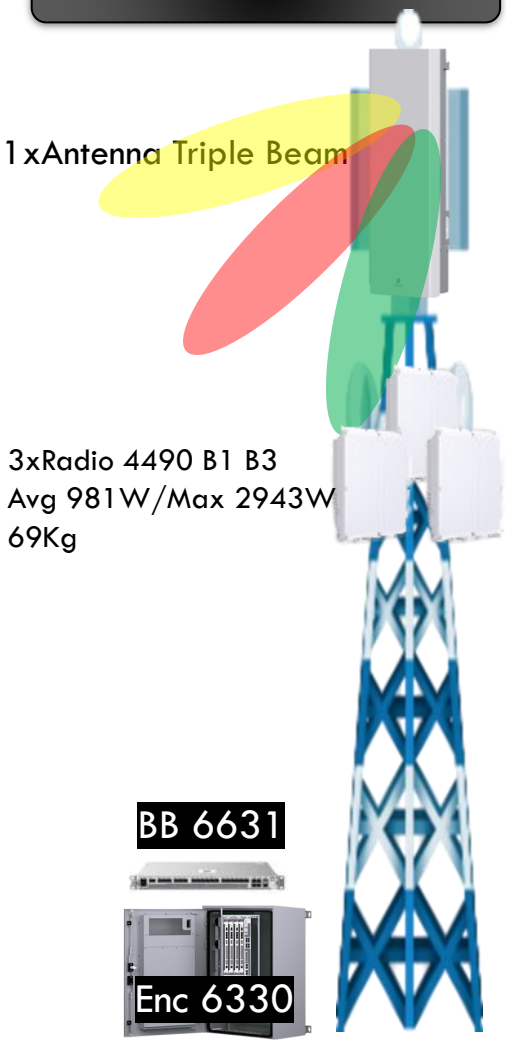
1x Radio 4490 B1 B3
1x Antenna Twin Beam

3x Radio 4490 B1 B3
1xAntenna Triple Beam

Radio 4490 1800+2100 MHz 2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VD

Antenna Triple Beam 900+1800+2100 +2300MHz 2G/4G/5G/Nb-IoT

Site Configuration



CAPACITY OPPORTUNISTIC:
OVER QUOTA SOLUTION

Telkomsel

Background

RCI or High Utilization with tower over quota , need capacity upgrade with ultimate hardware simplification.

Objective

- Add New NE to ease capacity congestion
- Boost Productivity & Revenue
- Optimize existing Infra

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

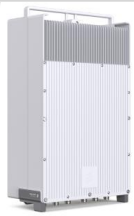
Solution

- Simplified Radio Solution

Scenario

Modernization Existing Module

3x RRUS12 B8
1893 W

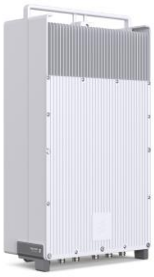


1x Radio 6626 B8 B28
810 W

3x RRUS12 B1+B3
3747 W

3x Radio 4490 B1 B3
2874 W

Radio 4490
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x100 Watt
4T4R
23kg | -48 VDC



Radio 6626
700+900 MHz
2G/4G/5G/Nb-IoT
6x120 Watt
3 x 2T2R
45kg | -48 VDC

Site Configuration

3xAntenna 12 Port
132Kg

3x AIR 3219 B40
Avg 906W/Max 1446W
51Kg

3xRadio 4490 B1 B3
Avg 981W/Max 2943W
69Kg

1x Radio 6626 B8 B28
Avg 400W/Max 810W
45Kg

BB 6631
BB 6651

Enc 6330

B2B CORPORATE USE CASES:
PRIVATE NETWORK ENTERPRISE

Telkomsel

Background

B2B Enterprise scenario with traffic offload between public network (MBB) & private network at Industrial park.

Objective

- Improve latency for private network
- Easy & Fast deployment
- Cost effective private network solution

Transport & Power & Infra

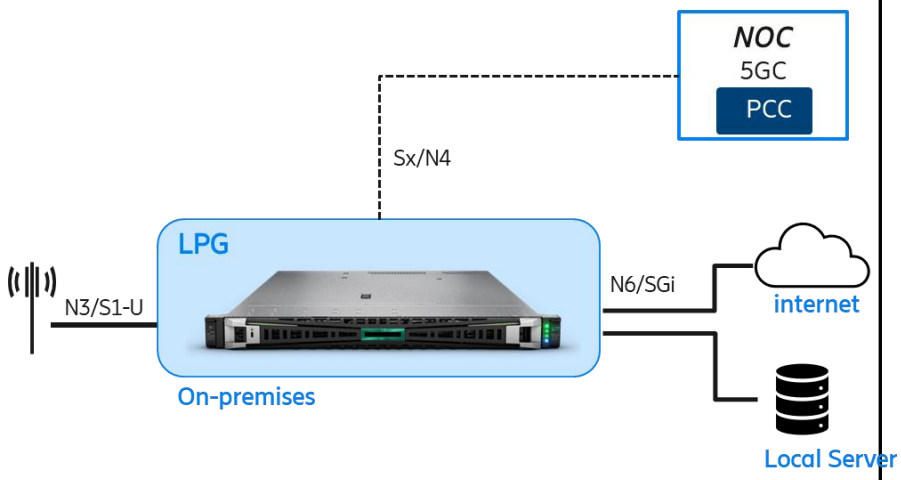
- Fiber Optic/Radio IP available
- PLN available
- Tower height 30 meter or more

Solution

- Light private network solution

Scenario

Private Network Solution



Solution

Single Server Dual Mode 5GC User Plane (Cloud Native Function). Support 4G, 5G NSA and 5G SA.

Centralized Control Plane in CSP NOC.

Benefit

Simpler, faster and more cost-effective footprint on-premises.

RAN Configuration

SUPPORT EVENT:
SMALL EVENT

Telkomsel

Background

Immediate support for event with small capacity requirement.

Objective

- Fast solution deployment for event support
- Solve coverage issue
- Increase productivity & revenue

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Mobile/Wall/Pole

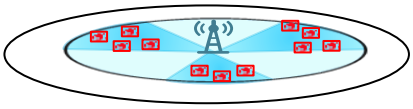
Solution

- Compact BTS solution

Scenario

Street Macro L1800+L2100

1 Site Radius < 1km



B2B Corporate

Sub Urban

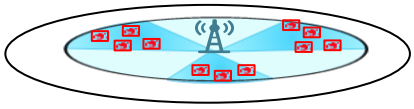
- Mid Population Area
- Estimated user population <1000

LTE 1800	3 x20MHz
LTE 2100	3 x20MHz (HW)
DCS 1800	3 x3Trx

AIR 4455
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x40 Watt
4T4R
27kg | -48 VDC

RMA L2300

2 Site Radius < 1km



B2B Corporate

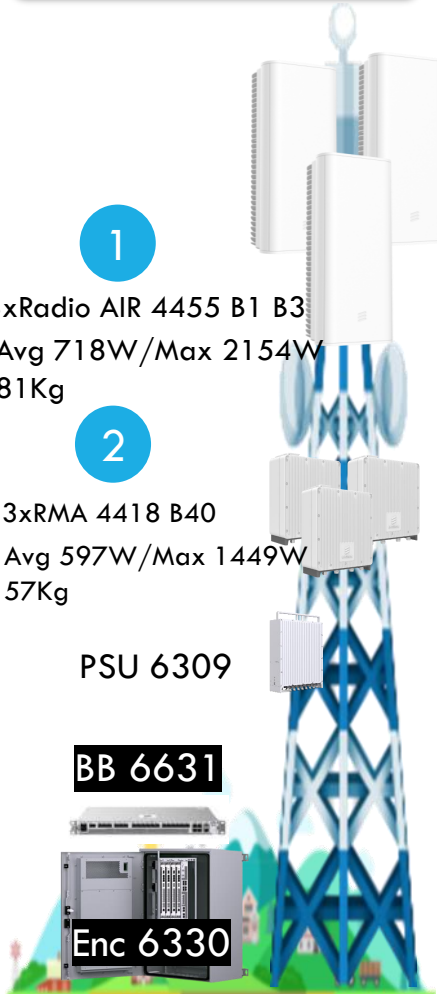
Sub Urban

- Mid Population Area
- Estimated user population <1000

LTE 2300	3 x50MHz

Radio 4418+RMA
2300 MHz
4G/5G/Nb-IoT
4x40 Watt
4T4R
19kg | AC/-48 VDC

Site Configuration



SUPPORT EVENT:
MID EVENT

Telkomsel

Background

Immediate support for event with medium capacity requirement.

Objective

- Fast solution deployment for event support
- Solve coverage issue
- Increase productivity & revenue

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Mobile/Wall/Pole

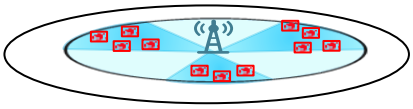
Solution

- Compact BTS solution

Scenario

Street Macro L1800+L2100

1 Site Radius < 1km



B2B Corporate

Sub Urban

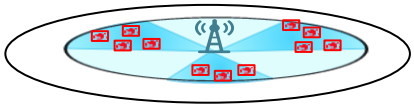
- Mid Population Area
- Estimated user population <1000

LTE 1800	**3 x20MHz**
LTE 2100	3 x20MHz (HW)

AIR 4455
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x40 Watt
4T4R
27kg | -48 VDC

RMA L2300

2 Site Radius < 1km



B2B Corporate

Sub Urban

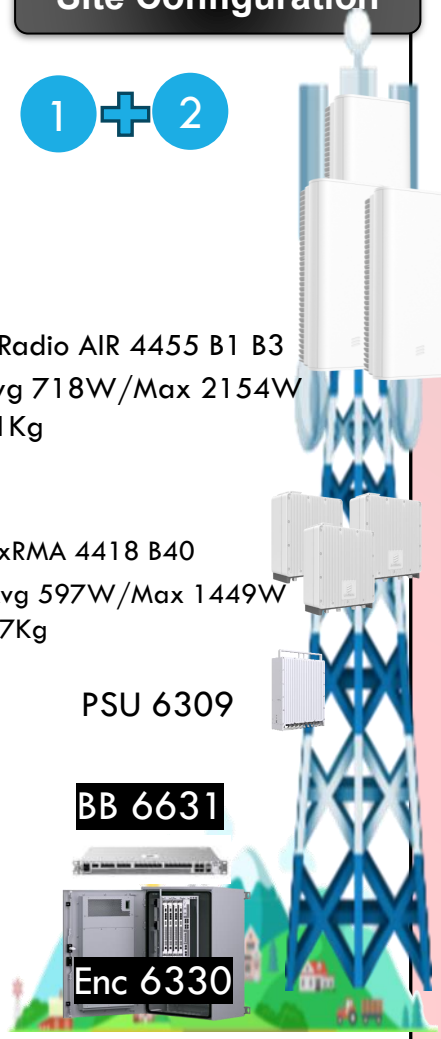
- Mid Population Area
- Estimated user population <1000

LTE 2300	**3 x50MHz**

Radio 4418+RMA
2300 MHz
4G/5G/Nb-IoT
4x40 Watt
4T4R
19kg | AC/-48 VDC

Site Configuration

1 + 2



3xRadio AIR 4455 B1 B3
Avg 718W/Max 2154W
81Kg

3xRMA 4418 B40
Avg 597W/Max 1449W
57Kg

PSU 6309

SUPPORT EVENT:
LARGE EVENT

Telkomsel

Background

Immediate support for event with high capacity requirement.

Objective

- Fast solution deployment for event support
- Solve coverage issue & high capacity requirement
- Increase productivity & revenue

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Combat BTS

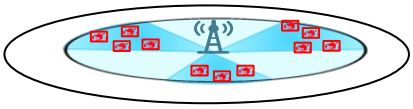
Solution

- Multiband Solution

Scenario

Street Macro L1800+L2100

1 Site Radius < 1km



B2B Corporate

Sub Urban

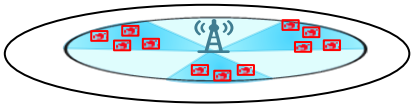
- Mid Population Area
- Estimated user population <1000

LTE 1800	**3 x20MHz**
LTE 2100	3 x20MHz (HW)
DCS 1800	3 x3Trx

AIR 4455
1800+2100 MHz
2G/4G/5G/Nb-IoT
4x40 Watt
4T4R
27kg | -48 VDC

Massive MIMO L2300

Site Radius < 2km



B2B Corporate

Sub Urban

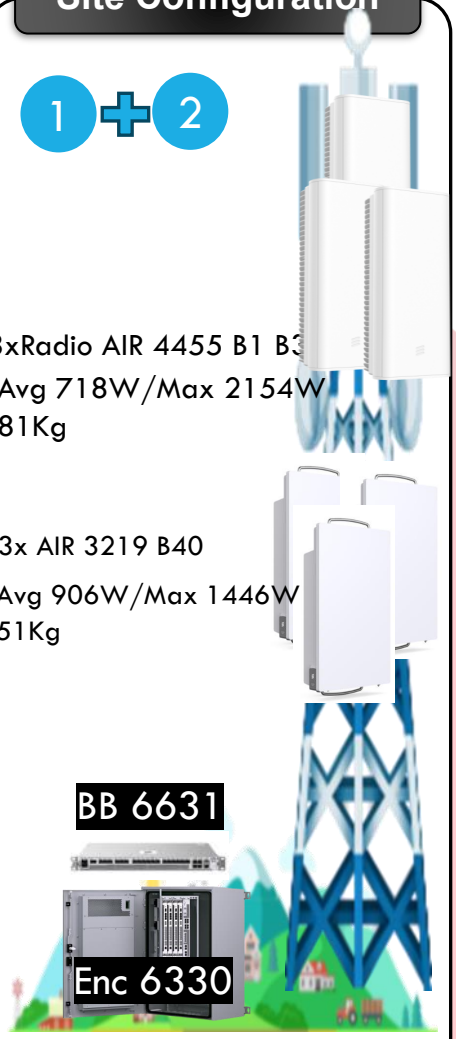
- High Population Area
- Estimated user population > 3000

LTE 2300	**3 x50MHz**

AIR 3219
2300 MHz
4G/5G/Nb-IoT
240 Watt
32T32R
27kg | -48 VDC

Site Configuration

1 + 2



SUPPORT EVENT:
INDOOR EVENT

Telkomsel

Background

Immediate support for event with indoor scenario.

Objective

- Fast solution deployment for indoor event support
- Solve indoor coverage issue
- Increase productivity & revenue

Transport & Power & Infra

- Fiber Optic/Radio IP available
- PLN available
- Wall/Pole

Solution

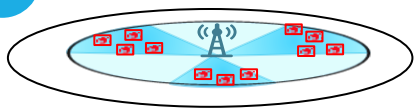
- Indoor Coverage Solution

Scenario

Micro L1800+L2100

Site Radius < 500m

1



B2B Corporate

Dense Urban

- Mid Population Area
- Estimated user population <1000

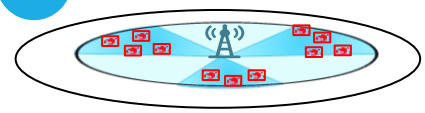
LTE 1800	**1 x20MHz**
LTE 2100	1 x20MHz

Radio 4402 B1
Radio 4402 B3
1800 MHz
4G/5G/Nb-IoT
4x5 Watt
4T4R
10kg | -48 VDC

Micro L2300

Site Radius < 500m

2



B2B Corporate

Dense Urban

- High Population Area
- Estimated user population <1000

LTE 2300	**1 x50MHz**

Radio 4408
2300 MHz
4G/5G/Nb-IoT
4x5 Watt
4T4R
10kg | -48 VDC

Site Configuration

RAN PRODUCT CATALOGUE

Product Radio FDD

Radio 4499 B8+B28:

- Dual Band 4T4R Radio - 4x 80
- Band 8 and 28
- Support up to 35 MHz IBW, 8 carriers GSM, 8 carriers WCDMA, and 4 carriers LTE in MIMO
- CPRI: 2x 2.5/4.9/9.8/10.1 Gbps
- 32 kg
- Power Supply : DC

Radio 4480 B1+B3:

- New Dual Band 4T4R Radio Band B1+B3 4x100W
- G, W, L, NR, NB-IoT with GSM in mixed mode
- 2x 2.5/4.9/9.8/10.1 Gbps CPRI
- 26.5 liter 32 kg
- Power Supply : DC

AIR 4455 B1+B3

- Dual band 4T4R B1+B3, 4x40W
- Antenna Gain: > 14.0 dBi
- Standards: G/W/L/NR/NB-IoT
- 2x 2.5/4.9/9.8/10.1 Gbps CPRI
- Power supply: AC
- Dimension: ~765x262x180 mm
- Weight: ~27 kg (without bracket)

Radio 4490 B1+B3:

- New Dual Band 4T4R Radio Band B1+B3 4x100W
- G, W, L, NR, NB-IoT with GSM in mixed mode
- 2x 10.1/25 Gbps CPRI
- 22.8 kg
- Power Supply : DC

Radio 6626 B8+B28

- New Dual Band 3x2T2R 6x120W
- G, W, L, NR, NB-IoT with GSM in mixed mode
- 2x 10.1/25 Gbps CPRI
- 45 kg
- Power Supply : DC

Radio 4466 B1+B3+B40

- New Triple Band 4T4R 4x180W
- G, W, L, NR, NB-IoT with GSM in mixed mode
- 2x 10.1/25 Gbps CPRI
- 45 kg
- Power Supply : DC

RAN PRODUCT CATALOGUE

Product Radio TDD

Radio 4418:
- 4T4R Radio-4x40W
- LTE TDD, NR
- Support split mode (2 x 2T2R)
- Up to 100MHz IBW TDD, Up to 6 LTE carriers
- 2x 2.5/5/9.8/ 10.1Gbps CPRI
- 16 liter, 17kg

Radio 8863:
- 8T38R Radio-8x50W
- LTE TDD, NR
- Support split mode (4x2T2R, 2x4T4R)
- Up to 100MHz IBW TDD,
- 2x 10.1/25Gbps CPRI
- 22.6 liter, 24.9kg

AIR3239:
- 32T32R Antenna Integrated Radio
- 100W
- LTE TDD, NR
- Up to 100MHz IBW TDD, Up to 3 LTE carriers
- 2x 10.1Gbps CPRI
- 30.5kg

AIR3219:
- 32T32R Antenna Integrated Radio
- 240W
- LTE TDD, NR
- Up to 100MHz IBW TDD, Up to 3 LTE carriers
- 2x 25Gbps CPRI
- 27kg

Radio 4466 B1+B3+B40
- New Triple Band 4T4R 4x180W
- G, W, L, NR, NB-IoT with GSM in mixed mode
- 2x 10.1/25 Gbps CPRI
- 45 kg
- Power Supply : DC

Radio 4408 B40
- New 4T4R Micro Radio Band B40 4x5W
- L, NR, NB-IoT
- 2x 10.1 CPRI
- 10 kg
- Power Supply : AC/DC

AIR6419:
- 64T64R Antenna Integrated Radio
- 320
- LTE TDD, NR
- Up to 100MHz IBW TDD, Up to 3 LTE carriers
- 2x 25Gbps CPRI
- 30kg

RAN PRODUCT CATALOGUE

Product Enclosure and Baseband

Baseband Indoor

BB5216:
- SDR

BB6630:
- SDR, NR, Support **Octa-Mode**

Outdoor Baseband

BB6318
- SDR, Legacy+NR
- Outdoor

New Generation

BB6631
- SDR,Legacy and NR
- **Triple Mode 2G+LTE+NR**

RP6651
- SDR
- **High Capacity LTE+NR**

Enclosure Outdoor

Enclosure 6330
- Capacity 3xBB + Transport
- 70Kg, DC -48V
- Baseframe or Pole

Telkomsel

RAN PRODUCT CATALOGUE

- FNI Not Yet
- FNI On Going
- FNI Finalization
- FNI Done

Upcoming Product (per August 2025)

Radio 4402

Radio 4402

Radio 4418

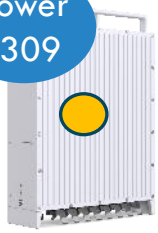
Radio 4402 B1
- New 4T4R Micro Radio Band B1 4x5W
- L, NR, NB-IoT
- 2x 10.1 CPRI
- 10 kg
- Power Supply : AC/DC

Radio 4402 B3
- New 4T4R Micro Radio Band B3 4x5W
- L, NR, NB-IoT
- 2x 10.1 CPRI
- 10 kg
- Power Supply : AC/DC

Radio 4418:
- 4T4R Radio-4x40W
- LTE TDD, NR
- Support split mode (2 x 2T2R)
- Up to 100MHz IBW TDD, Up to 6 LTE carriers
- 2x 2.5/5/9.8/ 10.1Gbps CPRI
- 19kg

Radio 4490

Power 6309



RP 6339

Radio 4490 B1+B3:
- New Dual Band 4T4R Radio Band B1+B3 4x100W
- G, W, L, NR, NB-IoT with GSM in mixed mode
- 2x 10.1/25 Gbps CPRI
- 25 kg
- Power Supply : DC

PSU AC 6309:
- Zero Footprint
- 4 x DC Output 30A
- 12 Kg
- Convert AC to DC

RP 6339 Outdoor Baseband
- SDR, LTE+NR
- Outdoor,10.5 Kg
- Power AC or DC

AIR 3285

AIR 3285 FDD Massive MIMO
- 1800+2100 MHz
- 4G/5G
- 320 Watt
- 32T32R
- 30kg | -48 VDC

Radio 6626

Radio 6626 B1 B3
- 1800+2100 MHz
- 2G/4G/5G/Nb-IoT
- 6x120 Watt
- 3 x 2T2R
- 45kg | -48 VDC